



Overfitting: When AI Memorises Instead of Learns



AI & Digital





OVERFITTING: WHEN AI MEMORISES INSTEAD OF LEARNS

Learning vs Memorising in Artificial Intelligence



What is AI?



AI helps us in games

AI helps us in phones

AI helps us in cars

AI learns patterns to solve problems

AI is used in many places!

AI is like a smart robot brain that learns from information. It helps us in many ways every day!

Learning vs Memorising

Learning:

- Understands the meaning behind information
- Can solve problems it has never seen before
- Adapts and grows smarter over time
- Like a student who understands a story, not just memorises words



The Key Difference

Memorising:

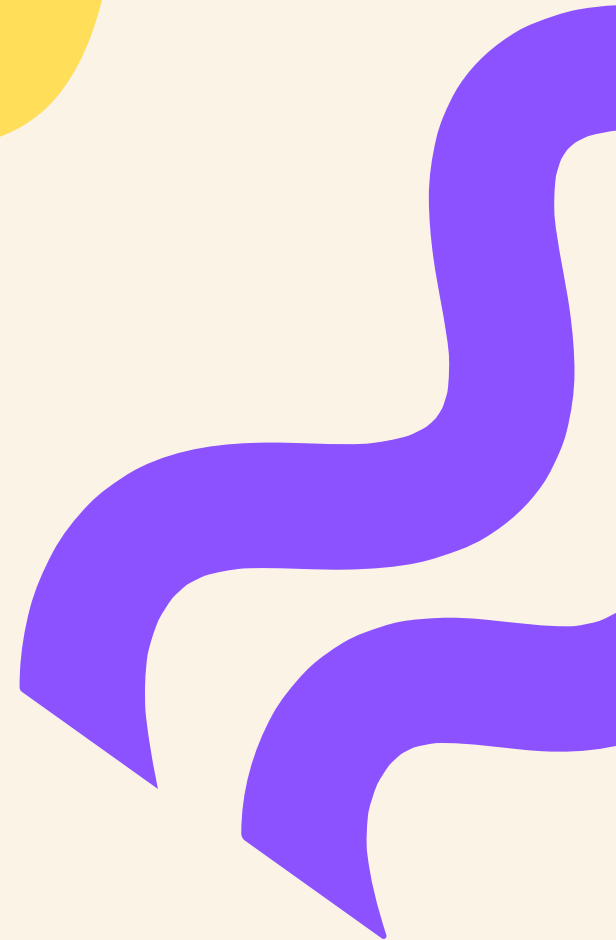
- Just repeats exactly what it saw before
- Cannot handle new or different situations
- Gets confused when things change
- Like a student who only remembers answers by heart

WHY DO AIS SOMETIMES MEMORISE?

AI sometimes memorises too much from the information it's trained on. Think of it like a student who only remembers answers by heart but can't solve new questions you give them!

Memorising = Just repeating exact answers from before

Learning = Understanding how to solve new problems



WHAT IS OVERFITTING?



**Overfitting means AI
memorises the training data too
perfectly instead of learning.**

It's like trying to remember every single question on a test by heart, but then getting stuck when you see a new question!

Example of Overfitting



AI memorises exact cat pictures

Can't recognise a different cat

Works only on training data

Gets confused with small changes

Fails on new examples

If the AI only remembers exact pictures, it can't recognise a new cat! This is overfitting in action.



Why is Overfitting a Problem?

AI becomes less smart and helpful

Makes mistakes when things change

Can't work with new information

Struggles when problems are different

Less useful for real-world tasks

How to Spot Overfitting

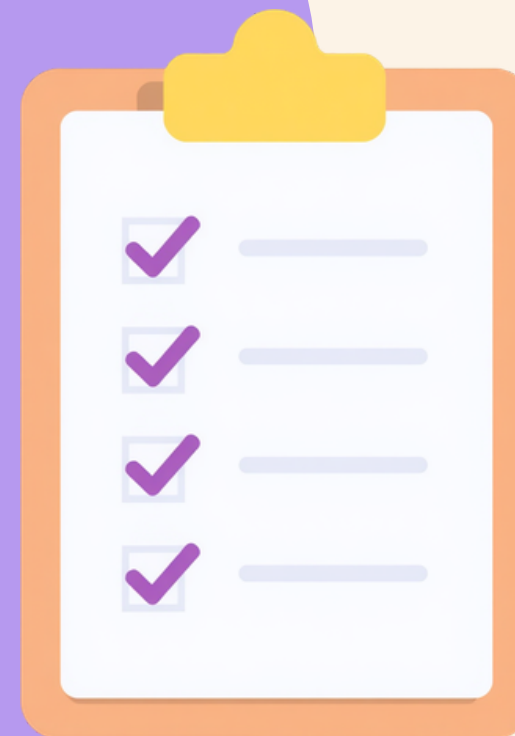
AI performs perfectly on training data but poorly on new data

AI's answers are too specific, not general

AI gets confused with small changes

AI doesn't adapt well to new situations

Like a student who only knows one way to solve problems



Thank You!

Any Questions?

Keep curious and keep learning about AI!

